HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

Worked Examples

ADVANCED TOPICS IN ENGINEERING PHYSICS

'Plasma and its applications'

Worked example 1

Task

- For surface cleaning purposes, the high voltage electrode of a DC-DBD plasma source, which is operated at atmospheric pressure, is located above a work piece made of pure aluminium. This work piece acts as counter electrode where the plasma discharge thickness is 2 mm. The aim is to remove a monolayer of grease (consisting of -C-H- single bonds) from the work piece via electron impact-induced dissociation.

    a) Estimate the required voltage which has to be applied to the electrode for obtaining such cleaning.

    b) Which plasma process gas should be used for the above-described application?

Worked example 2

Task

- In a high-power plasma generator, which can be described as a plane-parallel capacitor, the voltage applied to the electrodes made of copper is 10 kV. The distance between the electrodes is 1 mm. One copper electrode of this setup is irradiated by incident cosmic irradiation at a wavelength of 250 nm. The work function of the copper electrode material is 5.0 eV.

    - Is an electron released from the electrode material by the incident irradiation in the given case?

Worked example 3

Task

- A high-pressure LTE plasma is ignited between two plane and parallel electrodes where the discharge gap thickness is 1 mm. The process gas in the discharge gap is argon, the total particle density is $10^{30}/m^3$.

  a) Determine the required temperature of this plasma for obtaining ionisation of the process gas via electron impact-induced ionisation neglecting any acceleration of electrons by an electric field.

  b) Determine the required voltage that should be applied to the electrodes for obtaining ionisation of the process gas via electron impact-induced ionisation for the plasma at a temperature of 10,000K, however neglecting any temperature-driven acceleration of electrons.

Worked example 4

Task

- A plasma source consists of two plane-parallel electrodes with an overall discharge gap of 18 mm and is operated at atmospheric pressure. The applied alternating voltage amounts to 11 kV. Via incident cosmic irradiation, an electron is generated in the gas-filled discharge gap at a distance of 100 µm from the high-voltage electrode.

    a) Which high-voltage pulse frequency should be applied to ‘trap’ the electron between the electrodes…so that it does not impact on an electrode surface over a longer operation time of some minutes?

    b) Determine both the velocity and kinetic energy of the electron during oscillation as resulting from a).